🤖 Chatbot Documentation

🔍 Overview

The chatbot is a rule-based financial assistant developed using Python. It uses structured financial data from Microsoft, Apple, and Tesla to answer predefined queries related to revenue, growth, net income, and profitability. The chatbot provides responses by combining numerical data with simple financial insights.

⚙️ How the Chatbot Works

- The chatbot accepts user input as a text query.
- The query is converted to lowercase for consistent matching.
- It uses keyword-based conditions (if-else logic) to identify the company (Microsoft, Apple, Tesla) and the type of query (revenue, growth, income, profit).
- Relevant financial data is retrieved from a pandas DataFrame.
- A response is generated that includes both the financial value and a brief interpretation.

💬 Predefined Queries Supported

The chatbot can respond to the following types of queries:

- **Microsoft**
  - Revenue (e.g., “Microsoft revenue”)
  - Growth (e.g., “Microsoft growth”)
  - Profit margin (e.g., “Microsoft profit”)
- **Apple**
  - Revenue (e.g., “Apple revenue”)
  - Net income (e.g., “Apple net income”)
  - Profit margin (e.g., “Apple profit”)
- **Tesla**
  - Revenue (e.g., “Tesla revenue”)
  - Growth (e.g., “Tesla growth”)

    - Overall performance (default Tesla response)
- **Comparison**
    - Highest profit margin (e.g., “Which company has the highest profit margin?”)

⚠️ Limitations

- The chatbot only supports **predefined queries** and cannot handle complex or natural language variations.
- It uses **rule-based logic** and does not include advanced AI or NLP capabilities.
- Responses are limited to **fixed financial data (primarily 2025 values)**.
- The chatbot only covers **three companies** and a limited set of financial metrics.

🚀 Conclusion

This chatbot demonstrates how financial data can be integrated into a simple conversational interface. While basic, it provides a foundation for more advanced AI-driven financial analysis tools.